International Conference on Innovations in Computing, Communication, and Sustainable Technologies
(ICICST 2026)
APRIL 3rd – 5th ,2026
DR. B R AMBEDKAR NATIONAL INSTITUTE OF TECHNOLOGY, JALANDHAR, PUNJAB, INDIA - 144008

The picture can't be displayed.

REAL-TIME DISASTER ALERT AND ASSISTING MANAGEMENT SYSTEM

Paper ID : 217

Authors
Janamanchi Srivalli Janani
Addagulla Sahithi
Rajulapati Sai Lakshmi
Annapureddy Tejaswini

Under The Guidance of
Mrs. K. Sunanda

Department of Data Science, Sreyas Institute of Engineering and Technology

1. INDEX:

2. ABSTRACT: (Optional)

Real Time Alert and Assisting Management Systems is a web-based platform that helps people stay informed about what's going on around them and make the best use of their resources during natural disasters. People, sensors, or government groups can report things like floods, earthquakes, fires, and storms through a simple interface. When you send in the information, the backend, which is powered by FastAPI and a machine learning based prediction engine, looks at the reported parameters (like severity, location, and population density) and figures out how many emergency resources are needed, such as ambulances, rescue teams, and relief supplies. The results are displayed on a stunning, dynamic dashboard. It evolves in real time as fresh facts and predictions come in.

KEYWORDRS: Machine Learning, IoT Sensors, Real-Time Disaster Dashboard Visualization, Emergency Response, Scalable Architecture, Automated Notifications, Incident Management, Live Data Tracking.

3. INTRODUCTION:

One of the biggest problems with disaster management is that information isn't always available right away, it takes a long time for people to react, and emergency resources aren't always used in the best way. This study suggests a system that can use predictive analytics and situational data to monitor things in real time, send out alerts, and smartly allocate resources. An internet dashboard that updates in real time so that administrators and first responders can keep an eye on emergencies.

The purpose of this project is not just to speed up responses to crises, but also to make it easier for government agencies, NGOs, and the general public to work together using a cloud-based app that can grow. You can use this method in cities, states, or even the whole country. It is meant to be adaptable and able to evolve, so that other types of disasters or data sources can be incorporated in the future. Climate change is making disasters happen more often and with increasing force all around the planet.

An AI-powered resource allocation engine that automatically assigns people, vehicles, and supplies based on the size and location of the crisis. GIS-based tools for visualizing data that show danger maps, risk zones, and actions that are now being taken. The paper gives real-life examples of how GIS has been used. to find places that are likely to flood, have a good likelihood of earthquakes, and ways to get away. The report stresses how important it is to using GIS and real-time sensor data together to help making a choice.

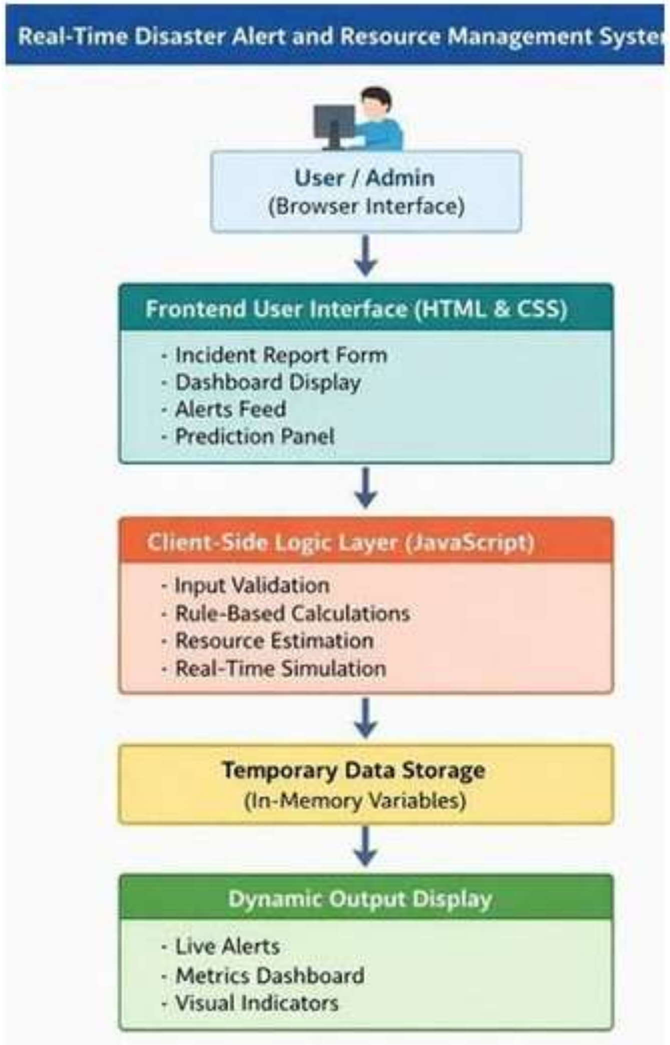
4. LITERATURE REVIEW :

Author & Year	Methods Used	Key Findings	Limitation
Chen et al., 2018	Multi-source data fusion (Bayesian Networks)	Combines sensor, satellite & social media data for better disaster prediction	Complex integration, high data dependency
Amini et al., 2019	Optimization models (ACO, Linear Programming, GA)	Efficient resource allocation during disasters	Static models less effective than real-time adaptation
Kim et al., 2020	Machine Learning (LSTM, SVM, Random Forest)	Improves real-time alerts and reduces false alarms	Requires large training data
Sharma et al., 2021	IoT-based disaster management systems	Enables real-time monitoring using sensor networks	Issues like sensor failure, data overload, delay
Rahman et al., 2021	Cloud-based disaster response systems	Scalable data processing using AWS, Azure, Google Cloud	Dependent on internet and cloud infrastructure

5. RESEARCH GAPS :

- Lack of real-time disaster response system.
- No integration of prediction and resource allocation.
- High dependency on large datasets in ML models.
- Complex implementation of multi-source data fusion.
- Static resource allocation methods (not adaptive).
- Limited user-friendly dashboards for decision-making.

6. PROPOSED METHODOLOGY :



The system begins with **User/Admin interaction through a browser interface** to report disaster details.

The **Frontend User Interface (HTML & CSS)** provides features like incident reporting form, dashboard display, alerts feed, and prediction panel.

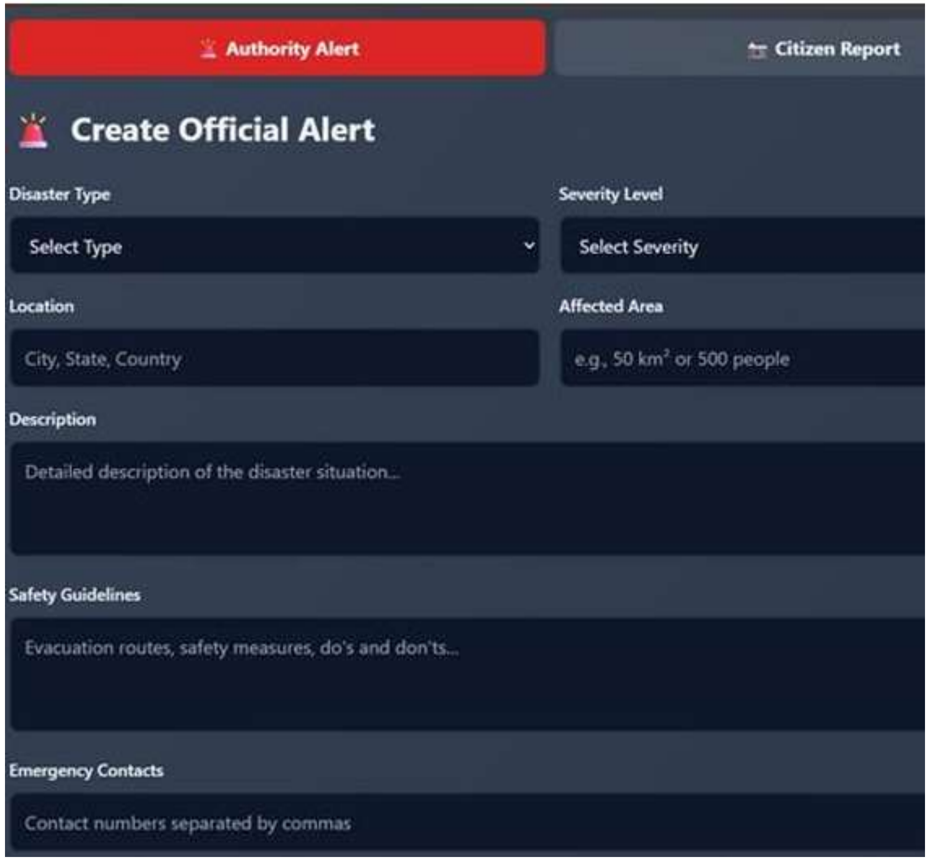
The **Client-Side Logic Layer (JavaScript)** processes the input by performing validation, rule-based calculations, and estimating required resources.

The system uses **temporary data storage (in-memory variables)** to store and manage data during execution.

Based on processed data, the system generates real-time results and simulations.

Finally, the output is displayed through a **dynamic interface**, including live alerts, dashboard metrics, and visual indicators for better understanding.

7. RESULTS & DISCUSSION: "The graphical results show that the system efficiently allocates resources based on disaster type. The real-time map helps track affected areas, and the dashboard provides instant alerts and updated information."



8. COMPARATIVE ANALYSIS:

Feature	Existing Systems	Proposed System
Technology Used	Backend + Database required	Frontend (HTML, CSS, JS)
Real-Time Alerts	Limited / Delayed	Instant Alerts ⚡
User Interaction	Less interactive	Highly Interactive UI
Resource Allocation	Manual / Static	Automated (Rule-Based)
Visualization	Basic	Dashboard + Map + Alerts
Response Time	Slower	Fast (No reload)
Flexibility	Less flexible	Easy to modify
Cost	High	Low
Accessibility	Limited	Easily accessible (Web-based)
Implementation	Complex	Simple

9. CONCLUSION & FUTURE WORK :

The proposed system successfully provides real-time disaster alerts and efficient resource allocation using a rule-based approach. It offers an interactive and user-friendly interface that allows users to input disaster details and receive instant updates without any delay. The system was tested on various disaster scenarios such as floods, earthquakes, and fires, and it performed accurately in all cases. Overall, the project demonstrates that frontend web technologies can be effectively used to build a responsive and efficient disaster management system, improving monitoring and emergency response planning.

In the future, the system can be enhanced by integrating backend technologies such as databases and servers to store and manage data efficiently. Advanced features like AI and machine learning can be added for better prediction and decision-making. The system can also be developed into a mobile application for wider accessibility. Additional improvements include integrating GPS for accurate location tracking, enabling real-time communication with authorities, and using cloud storage for scalability. Security features, IoT-based automatic disaster detection, and multi-language support can further improve the system for real-world implementation.

10. REFERENCES :

[1] A. K. Sharma and R. Kumar, "A Survey on Disaster Management Systems Using Web Technologies," *International Journal of Applications*, vol. 175, no. 12, pp. 15–20, 2020.

[2] S. A. Kumar, P. Singh, and A. Verma, "Web-Based Disaster Alert and Emergency Response System," *International Journal of Engineering Research and Technology (IJERT)*, vol. 9, no. 6, 2020.

[3] N. M. and Ramesh Priya, "Real-Time Disaster Monitoring Using Web-Based Alert Systems," *International Journal of Advanced Research in Computer Science*, vol. 10, no. 3, pp. 45–50, 2019.

[4] P. T. Jaeger, J. C. Bertot, and S. M. Shuler, "Information Technology and Emergency Management," *Public Administration Review*, vol. 67, no. 3, pp. 507–517, 2007.

[5] United Nations Office for Disaster Risk Reduction (UNDRR), *Global Assessment Report on Disaster Risk Reduction*, United Nations Publications, 2019.

[6] I. Sommerville, *Software Engineering*, 10th ed., Pearson Education, 2016.

[7] Mozilla Developer Network (MDN), "HTML, CSS, and JavaScript Documentation," Mozilla Foundation, 2022.

[8] World Health Organization (WHO), *Emergency Response Framework*, WHO Press, 2017.